

CSE 112:

Computer Organization
Sujoy Deb

Quiz II | Set-B

IIIT-Delhi, Winter 2025

Duration : 60 Minutes
Date: Mar 18, 2025
Points: 30

Guidelines

If you found any ambiguity in any of the questions or there appears to be a lack of information, then write an assumption on the answer sheet to explain your side interpretation of the problem and solve accordingly.

First Block Instructions	Description	Operation
addi rd,rs, imm[11:0]	Add immediate	$rd = rs + sext(imm[11:0])$
sub rd, rs1, rs2	Subtract registers	$rd = rs1 - rs2$
add rd, rs1, rs2	Add registers	$rd = rs1 + rs2$
and rd, rs1, rs2	Add registers	$rd = rs1 \& rs2$
xor rd, rs1, rs2	Perform XOR boolean operation	$rd = rs1 \text{ xor } rs2$
slt rd, rs1, rs2	Set Less than	$rd = 1$. If $signed(rs1) < signed(rs2)$ else $rd=0$
sltu rd,rs1,rs2	Set Less than unsigned	$rd = 1$. If $unsigned(rs1) < unsigned(rs2)$ else $rd=0$
beq rs1, rs2, imm[12:1]	Branch if equal	$PC = PC + sext((imm[12:1], 1'b0))$ if $rs1 == rs2$
beq x0,x0,#0	Branch if equal	HALT

Second Block Instructions	Description	Operation
bne rs1, rs2, imm[12:1]	Branch if not equal	$PC = PC + sext((imm[12:1], 1'b0))$ if $rs1 \neq rs2$
bge rs1,rs2, imm[12:1]	Branch if Greater than or Equal	$PC = PC + sext((imm[12:1], 1'b0))$ If $signed(rs1) > signed(rs2)$

[CO3] Problem I: Frequency of Operation

[01+02 Points]

We need to calculate the maximum frequency of operation for the two below-mentioned processors.

- A single stage processor with stage delay of 5ns.
- A Five stage processor having Fetch, Decode, Execute, Memory and Writeback and the stage delays are 7ns, 4ns, 5ns, 6ns and 8ns respectively.

[CO2] Problem II: Assembly Encoding

[10 Points]

We need to write an assembly code to produce the below-mentioned pattern.

Pattern

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[CO3] Problem III: MicroProcessor MicroArchitecture Design

[3x4 Points]

A task of designing a Microarchitecture for the Microprocessor having five stages (Fetch, Decode, Execute, memory, Writeback) is assigned to us. A set of stage components of different specifications are allocated to us. We need to design 4 microprocessors of the required specifications while minimizing the microprocessor cost by using the below-mentioned stage components.

Cost of any Stage component = 40 - delay_of_this_component.

Stage Components	(Name, Specification)			
Fetch Component	(F_A, 4ns)	(F_A, 6ns)	(F_A, 8ns)	(F_A, 9ns)
Decode Component	(D_A, 2ns)	(D_A, 10ns)	(D_A, 7ns)	(D_A, 5ns)
Execute Component	(E_A, 3ns)	(E_A, 18ns)	(E_A, 40ns)	(E_A, 22ns)
Memory Component	(M_A, 20ns)	(M_A, 2ns)	(M_A, 21ns)	(M_A, 32ns)
Writeback Component	(W_A, 3ns)	(W_A, 25ns)	(W_A, 6ns)	(W_A, 30ns)

Microprocessor	Kamet	Kabru	Kangto	Langpo
Specification (Maximum Operating Frequency)	200 MHz	50 MHz	40 MHz	25 MHz

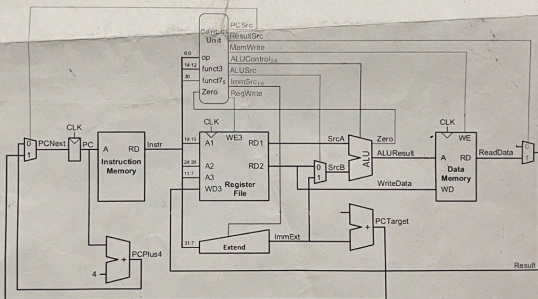
[CO3] Problem IV: Processor MicroArchitecture

[5 Points]

A Instructor intentionally removed some data paths from the below-mentioned Single Cycle Processor Microarchitecture to evaluate the student's understanding.

We as students need to answer which of the below-mentioned instructions will produce functionally correct results even after removal of datapaths.

Use **YES** or **NO** for correct and incorrect functionality respectively.



Single Cycle Processor [1]

Instruction	Status
lw rs2,imm[11:0](rs1)	Y
sub rd,rs1,rs2	N
sw rs2,imm[11:0](rs1)	Y
jalr rd,x6,offset[11:0]	N
beq rs1,rs2,imm[12:1]	N

[1] Harris, S., & Harris, D. (2021). *Digital Design and Computer Architecture, RISC-V Edition*. Morgan Kaufmann